AC9M6P02 • Year 6 Mathematics

Repeated Chance Experiments and Simulations

Compare observed frequency with expected probability and sample size

We are learning to…

Students calculate expected frequency, conduct physical and digital trials, compare relative frequencies across sample sizes and discuss why results vary while often becoming more stable.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

conduct repeated chance experiments and run simulations with an increasing number of trials; compare observed frequencies with expected frequencies and explain variation

Compare trial counts for a fair coin

trials	heads	relative frequency
10	**7**	**0.70**
50	**27**	0.54
200	103	0.515
1 000	498	0.498
expected		0.5

Larger samples often stabilise relative frequency near the model probability, but no trial count guarantees an exact match.

Audit a simulation and expected frequency

P(red)=0.3, 200 trials	physical trial	digital simulation	comparison	variation
expected red = 60	check device and consistent procedure	check probability settings and trial count	use relative frequency when totals differ	describe difference without changing data

A simulation is only as valid as its probability model and implementation. Verify settings before interpreting output.

Build and annotate the model

Represent repeated chance experiments and simulations and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Expected frequency treated as guaranteed:** It is a long-run prediction.
- **Raw counts compared across unequal totals:** Use proportions.
- **Unwanted outcomes deleted:** Record all valid trials.
- **Simulation settings assumed correct:** Audit the model.

Quick check

- Calculate expected count.
- Find relative frequency.
- Compare 10/50/200 trials.
- Explain variation.
- Check simulation settings.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.